HIGHER DIRECT IMAGES OF QUASI-COHERENT SHEAVES VANISH PAST DEGREE ONE FOR PROPER MAPS WITH AT-MOST-CURVE FIBRES, AND THE FIRST DIRECT IMAGE VANISHES FOR GLOBALLY GENERATED SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20970-20986 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20988-21015, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's sentence contains a typo (remains is the prove (4)); retained verbatim.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The dimension-bound on cohomological amplitude (Limits, lemma-higher-direct-images-zero-above-dimension-fibre), the chapter's lemma-h1-fibre-zero, quasi-coherence of higher direct images, and colimit commutation for H^1 on quasi-separated schemes.

Source reference. Original lemma 01XJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21015–21015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01XK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21015–21015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01FF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21015–21015. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20943–20949. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21015–21015. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes such that $\dim(X_y) \leq 1$ and $H^1(X_y, \mathcal{O}_{X_y}) = 0$ for all $y \in Y$. Let $\mathcal{F}$ be quasi-coherent on X. Then*

(1) *$R^p f_* \mathcal{F} = 0$ for $p > 1$, and*
(2) *$R^1 f_* \mathcal{F} = 0$ if there is a surjection $f^* \mathcal{G} \to \mathcal{F}$ with $\mathcal{G}$ quasi-coherent on Y.*

If Y is affine, then we also have

(3) *$H^p(X, \mathcal{F}) = 0$ for $p \notin \{0, 1\}$, and*
(4) *$H^1(X, \mathcal{F}) = 0$ if $\mathcal{F}$ is globally generated.*

Proof. The vanishing in (1) is Limits, Lemma 0E7D. To prove (2) we may work locally on Y and assume Y is affine. Then $R^1 f_* \mathcal{F}$ is the quasi-coherent module on Y associated to the module $H^1(X, \mathcal{F})$. Here we use that Y is affine, quasi-coherence of higher direct images (Cohomology of Schemes, Lemma 01XJ), and Cohomology of Schemes, Lemma 01XK. Since Y is affine, the quasi-coherent module $\mathcal{G}$ is globally generated, and hence so is $f^* \mathcal{G}$ and $\mathcal{F}$. In this way we see that (4) implies (2). Part (3) follows from (1) as well as the remarks on quasi-coherence of direct images just made. Thus all that remains is the prove (4). If $\mathcal{F}$ is globally generated, then there is a surjection $\bigoplus_{i \in I} \mathcal{O}_X \to \mathcal{F}$. By part (1) and the long exact sequence of cohomology this induces a surjection on H^1. Since $H^1(X, \mathcal{O}_X) = 0$ because $R^1 f_* \mathcal{O}_X = 0$ by Lemma 0E7G, and since $H^1(X, -)$ commutes with direct sums (Cohomology, Lemma 01FF) we conclude. □